

A laminar neural mass model of the human somatosensory system: from thalamocortical dynamics to the EEG somatosensory evoked response

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04.08.2026

Abstract

[Placeholder] We present a laminar neural mass model of the human somatosensory system comprising thalamus (VPM, reticular nucleus, POm), Brodmann area 3b, and the primary and secondary somatosensory cortices (S1, S2), each resolved into cortical layers and into excitatory, PV, SST and VIP populations. Model dipoles are projected to the scalp through an individual forward model, allowing a direct comparison of simulated and measured somatosensory evoked potentials and of pre-stimulus spectral activity.

Introduction

[Placeholder]

Methods

Neural mass model of the somatosensory system

Model architecture

The model consists of 33 interacting neural populations distributed over four regions. The thalamus is represented by three populations: a first-order excitatory relay population (ventral posteromedial nucleus, VPM), an inhibitory population representing the reticular nucleus, and a higher-order excitatory population (posterior medial nucleus, POm). Brodmann area 3b is modelled as a single, layer-collapsed cortical column with four populations (excitatory pyramidal cells E, and the inhibitory interneuron classes PV, SST and VIP). The primary (S1) and secondary (S2) somatosensory cortices are each modelled with 13 populations: an excitatory, a PV and an SST population in each of the layers 2/3, 4, 5 and 6, plus a single VIP population in layer 2/3.

Each population i is described by its mean membrane potential V_i and its mean firing rate r_i . Every connection from a source population j to a target population i has its own postsynaptic potential state v_{ij} , so that excitatory and inhibitory input to the same target are filtered with different kinetics before being summed.

Population dynamics

Synaptic transmission is modelled by the convolution of the presynaptic firing rate with an alpha function, as in the classical Jansen–Rit formulation. In differential form, the postsynaptic potential v_{ij} evoked in population i by the firing rate r_j of population j obeys

$$\frac{d^2 v_{ij}}{dt^2} = \frac{H_{ij}}{\tau_{ij}} W_{ij} r_j - \frac{2}{\tau_{ij}} \frac{dv_{ij}}{dt} - \frac{1}{\tau_{ij}^2} v_{ij}, \quad (1)$$

where W_{ij} is the effective connectivity weight, H_{ij} the synaptic gain and τ_{ij} the synaptic time constant. The two input channels of the model, the external (stimulus) input I_{ext} and the background input I_{bg} , are filtered by the same synaptic kernel,

$$\frac{d^2 v_i^{\text{ext}}}{dt^2} = \frac{H_i}{\tau_i} W_i^{\text{ext}} I_{\text{ext}} - \frac{2}{\tau_i} \frac{dv_i^{\text{ext}}}{dt} - \frac{1}{\tau_i^2} v_i^{\text{ext}}, \quad (2)$$

$$\frac{d^2 v_i^{\text{bg}}}{dt^2} = \frac{H_i}{\tau_i} W_i^{\text{bg}} I_{\text{bg}} - \frac{2}{\tau_i} \frac{dv_i^{\text{bg}}}{dt} - \frac{1}{\tau_i^2} v_i^{\text{bg}}. \quad (3)$$

The total membrane potential of a population is the sum of all postsynaptic contributions it receives, together with the two input channels,

$$V_i = \sum_j v_{ij} + v_i^{\text{ext}} + v_i^{\text{bg}}, \quad (4)$$

and is converted into a mean firing rate by a sigmoidal transfer function,

$$r_i = \frac{r_i^{\text{max}}}{1 + \exp[a_i(\theta_i - V_i)]}. \quad (5)$$

The parameters of Eq. (5) are cell-type specific: excitatory populations use $a = 0.128 \text{ mV}^{-1}$, $\theta = 32.1 \text{ mV}$ and $r^{\text{max}} = 31.4 \text{ s}^{-1}$; PV populations $a = 0.142 \text{ mV}^{-1}$, $\theta = 40.0 \text{ mV}$ and $r^{\text{max}} = 166.8 \text{ s}^{-1}$; SST populations $a = 0.079 \text{ mV}^{-1}$, $\theta = 42.0 \text{ mV}$ and $r^{\text{max}} = 57.0 \text{ s}^{-1}$; and VIP populations $a = 0.070 \text{ mV}^{-1}$, $\theta = 37.9 \text{ mV}$ and $r^{\text{max}} = 38.5 \text{ s}^{-1}$. The synaptic time constant τ_{ij} is determined by the presynaptic cell type, with 6 ms for excitatory, 3 ms for PV, 20 ms for SST and 15 ms for VIP synapses. In the present implementation the synaptic gains H_{ij} are set to unity, so that the amplitude of each connection is carried entirely by the connectivity weight W_{ij} described below.

Connectivity

The connectivity weight of each projection is constructed from anatomical quantities as

$$W_{ij} = P_{ij} S_{ij} C_j, \quad (6)$$

that is, the probability P_{ij} that a neuron of population j contacts a neuron of population i , multiplied by the mean unitary postsynaptic potential amplitude S_{ij} of that connection and by the number of neurons C_j in the presynaptic population. Connection probabilities and unitary amplitudes for the intra-columnar connectivity of S1 and S2 were taken from published cell-type and layer resolved connectivity data; the connectivity of area 3b was obtained by collapsing the S1 matrix over layers, using cell-count weighted averages of both the source and the target populations.

The resulting weights are then scaled by a small number of free gain parameters. All excitatory columns of the matrix are multiplied by g_E and all inhibitory columns by $-g_I$, so that inhibitory projections enter Eq. (1) with a negative sign. The output of the thalamic populations is scaled separately, with g_E^{thal} for the VPM, $-g_I^{\text{thal}}$ for the reticular nucleus and g^{POm} for the POm. Inter-areal excitatory projections (area 3b $\leftrightarrow$ S1 $\leftrightarrow$ S2, feedforward from layers 2/3 and 5 onto layers 2/3, 4 and 5 of the receiving area, and the corresponding feedback projections) are scaled by a separate parameter g_{inter} , which allows the strength of the cortico-cortical hierarchy to be varied independently of the local circuitry. Corticothalamic projections originate from layer 6 (to VPM and reticular nucleus) and from layer 5 (to POm).

Conduction delays are implemented as additive offsets on the synaptic time constants of the corresponding projections: 5 ms for long-range cortico-cortical connections between area 3b, S1 and S2, and 3 ms for thalamocortical connections. When simulated responses are compared to the measured evoked potentials, an additional fixed delay of 20 ms accounts for the conduction time from the periphery to the thalamus, which is not part of the simulated network.

Inputs and background noise

The stimulus is delivered as an external input I_{ext} to the VPM population only, modelled as a rectangular pulse of a given amplitude and duration starting at the stimulus onset; all other populations receive $W_i^{\text{ext}} = 0$. All cortical populations additionally receive a background input I_{bg} that represents ongoing drive from areas outside the modelled network. To generate ongoing activity with a realistic spectrum, this background input carries independent colored noise for each population, implemented as an Ornstein–Uhlenbeck process with zero mean, stationary standard deviation σ_{bg} (in units of the mean background drive) and correlation time $\tau_{\text{bg}} = 16$ ms. The process is generated with the exact discrete update

$$\eta_i(t + \Delta t) = e^{-\Delta t/\tau_{\text{bg}}} \eta_i(t) + \sigma_{\text{bg}} \sqrt{1 - e^{-2\Delta t/\tau_{\text{bg}}}} \xi_i(t), \quad \xi_i(t) \sim \mathcal{N}(0, 1), \quad (7)$$

so that the noise statistics are independent of the integration step size, and it is initialised from its stationary distribution. Setting $\sigma_{\text{bg}} = 0$ recovers the deterministic model with constant background drive. Because the network needs time to settle into its stationary regime, the stimulus onset is placed at least 2 s after the start of the simulation whenever pre-stimulus spectral properties are analysed.

Numerical integration

The system of Eqs. (1)–(5) was integrated with a forward Euler scheme at a step size of $\Delta t = 1$ ms over a total simulation duration of 3 s. All state variables were initialised at zero. Table 1 summarises the symbols used above.

Table 1: Variables and parameters of the neural mass model.

Symbol	Description
v_{ij}	Postsynaptic potential in population i from population j
$v_i^{\text{ext}}, v_i^{\text{bg}}$	Postsynaptic potential from external and background input
V_i	Total membrane potential
r_i	Population firing rate
r_i^{max}	Maximum firing rate
a_i	Sigmoid slope
θ_i	Firing threshold
W_{ij}	Effective connectivity weight
H_{ij}	Synaptic gain
τ_{ij}	Synaptic time constant
P_{ij}	Connection probability from population j to population i
S_{ij}	Mean unitary postsynaptic potential amplitude
C_j	Number of neurons in population j
g_E, g_I	Coupling strength of excitatory and inhibitory populations
g_{inter}	Coupling strength of inter-areal projections
I_{ext}	External stimulus
I_{bg}	Background input
$\sigma_{\text{bg}}, \tau_{\text{bg}}$	Standard deviation and correlation time of the background noise

Forward modelling of the EEG

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Model fitting

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Results

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Discussion

Limitations

Implications

Conclusion

References

References